

NLP in learner corpus research

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Topics to cover

- Participation opportunity
- What is learner corpus research
- How can LCR be connected to NLP
- Presentation: Kyle & Eguchi (2024)

Participation opportunity

Grading

- (4) Participation 10%
 - Completing ad-hoc events that the instructor announces in class 2%
 - Active participation in Presentation 4%
 - Active participation in midway/final presentation 4%

GitHub Issue

A **GitHub Issue** is a message attached to a repository.

You can use an issue to:

- ask a question
- report a problem or error (similar to 'help ticket')
- suggest an improvement (lecture notes, tutorials)

I will use the issues you raise in our course repository

(https://github.com/hksung/Fall26_SLS775) to fix or improve course materials, and your issue activity will also be reflected in your participation grade (2%).

What is learner corpus research

Learner corpus research (LCR) (Granger, 2026)

Learner corpus research (LCR) uses learner corpora to study the characteristics and development of learner language.

Two main goals:

- **Theoretical:** Understand learner language and factors that shape it (e.g., L1, proficiency, context, task).
- **Pedagogical:** Inform teaching materials, assessment, and language-learning tools.

Overview of LCR

We will examine LCR from five perspectives:

1. Language data
2. Methodological approaches and research designs
3. Linguistic foci
4. Contributions to SLA
5. Contributions to L2 teaching and assessment

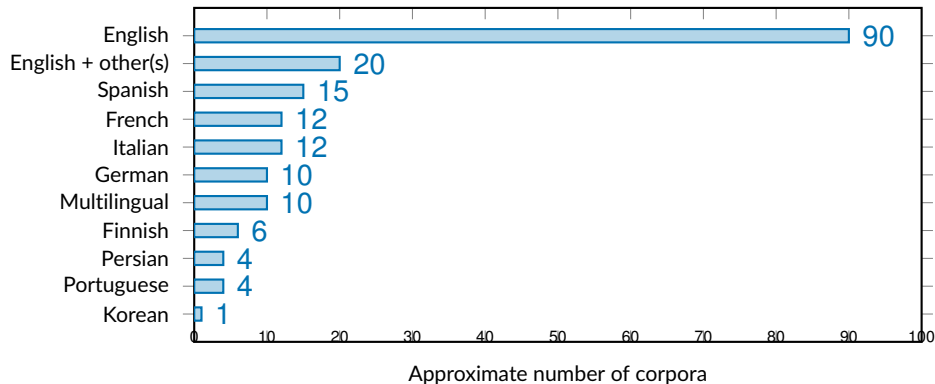
1. Language data: Language

- Early learner corpora focused mainly on L2 English
 - Learner corpora are now available for a much wider range of target languages
 - English, however, remains disproportionately represented

Resource: Learner Corpora around the World

<https://uclouvain.be/en/research-institutes/ilc/cecl/learner-corpora-around-the-world.html>

1. Language data: CECL (example)



1. Language data: raw vs. annotated corpora

Raw corpora

- Original learner texts or transcribed speech
- Often stored as .txt
- Metadata may be stored separately

Annotated corpora

- Learner data enriched with linguistic information
- May use formats (e.g., .xml - a markup language)
- Annotation depends on the research purpose

Example: MICASE

```
▼<SETTINGDESC>
  ▼<SETTING>
    <P> Honors Advising Office, Angell Hall </P>
  </SETTING>
</SETTINGDESC>
</PROFILEDESC>
</TEIHEADER>
▼<TEXT>
  ▼<BODY>
    ▼<U1 WHO="S1" NSS="NRN" ROLE="ST" SEX="F" AGE="4" RESTRICT="NONE" FLANG="EST">
      so. i see that you're from Hartland Michigan
      <U2 WHO="S2" NSS="NS" ROLE="JU" SEX="F" AGE="1" RESTRICT="NONE"> yes </U2>
      this is, right up the road
    </U1>
    ▼<U1 WHO="S2" NSS="NS" ROLE="JU" SEX="F" AGE="1" RESTRICT="NONE">
      mhm, like forty minutes from here
      <U2 WHO="S1" NSS="NRN" ROLE="ST" SEX="F" AGE="4" RESTRICT="NONE" FLANG="EST"> yeah </U2>
      mhm
    </U1>
    <U1 WHO="S1" NSS="NRN" ROLE="ST" SEX="F" AGE="4" RESTRICT="NONE" FLANG="EST"> okay, and uh, you say that you're interested in prebusiness and economics </U1>
    ▼<U1 WHO="S2" NSS="NS" ROLE="JU" SEX="F" AGE="1" RESTRICT="NONE">
      i was i don't think that i am anymore
      <EVENT DESC="LAUGH"/>
    </U1>
```

Michigan Corpus of Academic Spoken English

(MICASE, <https://quod.lib.umich.edu/cgi/c/corpus/corpus?c=micase;cc=micase;view=transcript;id=ADV700JU023>)

Example: MICASE

2.4 Speaker attributes

CATEGORY	CODE	DEFINITION/COMMENTS
GENDER		
FEMALE	F	
MALE	M	
AGE GROUP		
17 - 23	1	
24 - 30	2	
31 - 50	3	
51 and older	4	
ACADEMIC ROLE		
JUNIOR UNDERGRAD	JU	First and second year undergraduates
SENIOR UNDERGRAD	SU	Third year and above undergraduates
JUNIOR GRADUATE	JG	First and second year or Master's level graduate students
SENIOR GRADUATE	SG	Third year and above Ph.D. students
JUNIOR FACULTY	JF	Lecturers and Assistant Professors
SENIOR FACULTY	SF	Associate Professors and above
RESEARCHER	RE	Non-teaching researchers
POST-DOC FELLOW	PD	Post-doctoral research fellows
STAFF	ST	Non-teaching University employees
VISITOR/OTHER	VO	Non-University of Michigan affiliates

2. Methodological approaches and research designs

Contrastive Interlanguage Analysis

- Compares learner language with a reference variety (traditionally, “native” language) (Granger, 1996)
- CIA², a revised model of CIA that places greater emphasis on variation across both learner and reference varieties (Granger, 2015)
- Identifies features that are characteristic of learner language (vs. language from *reference corpora*)

2. Methodological approaches and research designs

Longitudinal design

- Collects repeated data from the same learners over time
- Provides more direct evidence of individual development
- Attrition and changing contexts/tasks

2. Methodological approaches and research designs

Cross-sectional design

- Compares different groups of learners at one point in time
- Often compares proficiency levels or instructional stages

Pseudo-longitudinal design

- Uses cross-sectional groups at different proficiency or instructional stages
- Treats group differences as an approximation of developmental change

3. Linguistic foci

Learner corpus research examines a wide range of linguistic phenomena, but some areas have received considerably more attention than others.

According to the *Learner Corpus Bibliography* (from UCLouvain):

- **Lexis:** ~65%
- **Grammar:** ~46%
- **Discourse:** ~30%
- **Pragmatics:** ~10%
- Categories are not mutually exclusive; one study may investigate multiple areas.

3. Linguistic foci: Phraseology (example)

Phraseology: the study of recurrent and conventional word combinations (e.g., collocations, lexical bundles, and other formulaic sequences).

Researchers may ask whether L2 learners:

- use a narrower range of word combinations?
- use collocations or lexical bundles differently from other groups?
- produce combinations that are grammatically possible but uncommon in the target register?
- anything else?

3. Linguistic foci: Written vs. Spoken

Most learner corpus research has focused on written language, particularly *academic* writing.

- Written texts are relatively easy to collect and store.
- They are easier to process and annotate than spoken interaction.
- Advanced academic writing has important pedagogical applications.

As a result, we know much more about L2 academic writing than about:

- conversational language
- informal registers
- interactional development
- lower-proficiency learner production

4. Contributions to SLA

Original goal: Use learner corpus evidence to contribute to SLA theory.

Early research was often:

- primarily *descriptive*
- focused on *frequencies* and group differences
- only loosely connected to theories of language acquisition

As a result, learner corpus findings initially had relatively limited influence in the broader SLA community (Tracy-Ventura & Paquot (2021)).

4. Contributions to SLA: Descriptive study (example)

e.g., We compare the use of *however* in academic writing produced by English learners and “native” English speakers.

	Learner essays	Native essays
<i>however</i> per 10,000 words	18.4	7.2

A descriptive study might conclude:

Learners use however more frequently than expert writers.

4. Contributions to SLA: Beyond descriptions

Beyond descriptions, they are useful for examining:

- what learners actually produce
- patterns across large and diverse learner populations
- variation across proficiency levels, L1s, tasks, and contexts
- developmental patterns in learner language

4. Contributions to SLA: (1) Generalizability

Large and diverse learner corpora can test whether findings from small samples generalize across learner populations.

Example: Murakami & Alexopoulou (2016)

- Examined the proposed universal order of acquisition of English morphemes
- Used large-scale learner data from learners with different L1 backgrounds
- Found that L1 substantially influenced acquisition patterns; Findings that appear “universal” in a few learner groups may not generalize to all L2 learners

See <https://doi.org/10.1017/S0272263115000352>

4. Contributions to SLA: (2) Linguistic Complexity

Learner corpus research has contributed to the operationalization of linguistic complexity.

- lexical complexity;
- syntactic / grammatical complexity;
- phraseological complexity.

We will return to linguistic complexity in more detail in Week 8.

4. Contributions to SLA: (3) Predictive Studies

Learner corpus studies can also ask which linguistic features are associated with learner outcomes (often related to assessment).

- Does X feature distinguish Y?

Table 4. Component 1: informational ($r = -.569$ with text readability).

Feature	Loading	Interpretation
Orthographic neighbors (Levenshtein Distance)	0.92	More lexical complexity
Number of suffixes (CW)	0.92	More morphological complexity
Word age of acquisition (Kuperman)	0.90	More lexical complexity
Number morphemes (CW)	0.90	More morphological complexity

See Crossley (2024), *Developing Linguistic Constructs of Text Readability Using Natural Language Processing*.

doi.org/10.1080/10888438.2024.2422365

5. Contributions to teaching and assessment

Learner corpus research has long aimed to make L2 teaching and assessment more responsive to learners' actual language use.

Learner corpus evidence can help identify:

- recurrent grammatical, lexical, and phraseological difficulties
- linguistic features associated with more effective/authentic target-language use

These findings can inform:

- pedagogical materials and classroom activities
- proficiency descriptions and assessment criteria

5. Teaching Applications: LEAD (example)

LEAD: Louvain English for Academic Purposes Dictionary

- A web-based academic writing resource informed by analyses of learner and expert academic writing.
- It helps learners identify academic words that are often (1) misused, (2) overused, or (3) underused.
- It also provides information about (1) collocations, (2) lexical bundles, or (3) common phraseological patterns.

Granger & Paquot (2015, 2022)

The Louvain EAP Dictionary

The LEAD content partially adapts to user's mother tongue and field of interest. Before using the dictionary, you should therefore select a discipline and specify your language background. We also ask you to tell us where you are from - we'd love to know more about our users !

Discipline

General EAP

Mother tongue

French

Country

United States

Continue

<https://leaddico.uclouvain.be/>

5. Teaching Applications: Data-Driven Learning

Data-Driven Learning (DDL) involves learners examining corpus examples and discovering linguistic patterns themselves.

- (+) Learners can observe patterns in authentic language data
- (-) Learner-corpus-based DDL shows promising results (Hartle, 2023), although its direct classroom use remains relatively limited.

How can LCR be connected to NLP

Overview

NLP uses *computational methods* to analyze human language automatically. These methods include:

- (build computational models)
- automatically annotate linguistic features
- extract and quantify linguistic information at scale
- search for and compare linguistic patterns (e.g., concordancers).

NLP pipeline



Common NLP (pre-)rocesses

- Tokenization
- Lemmatization
- POS tagging
- Parsing

Tokenization: English (example)

Tokenization divides text into units called **tokens**.

She walked to school.

Word-level:

She | walked | to | school | .

Morpheme-level:

she | walk | -ed | to | school | .

*What counts as a “token” depends on the language and the purpose of the analysis.

Tokenization: Korean

Tokenization becomes more complicated when **spaces do not correspond directly to linguistic units**.

저는 학교에 갑니다.

Space-delimited units:

저는 | 학교에 | 갑니다 | .

Morpheme-level units:

저 | 는 | 학교 | 에 | 가 | ㅂ니다 | .

Lemmatization

Lemmatization maps inflected forms to a shared dictionary form (**lemma**).

walks, walked, walking → **walk**

But lemmatization is **not always desirable**:

- combining forms may remove meaningful morphological information;
- its usefulness depends on the language and research question.

POS Tagging

Part-of-speech (POS) tagging assigns grammatical categories to individual tokens.

Coarse-grained

- NOUN
- VERB
- ADJ
- ADV

Fine-grained

- NN vs. NNS
- VB, VBD, VBG
- VBN, VBP, VBZ

Parsing

Parsing automatically analyzes the syntactic structure of sentences.

Constituency parsing

- groups words into phrases
- represents hierarchical structure
- e.g., S, NP, VP

Dependency parsing

- labels grammatical relations
- identifies heads and dependents
- e.g., subj, obj

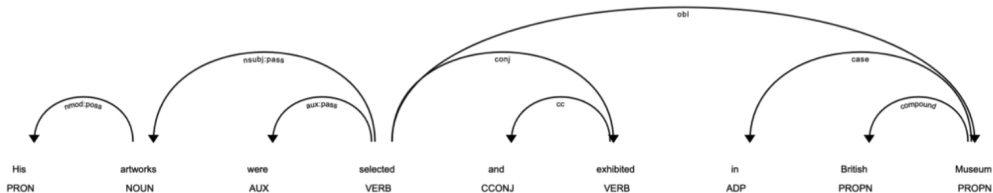


Figure 3: An example of dependency parse tree with Universal Dependencies, visualized by *displaCy*. Adapted from [Kyle and Sung \(2026\)](#)

NLP accuracy

Automatic annotation usually involves an *interdependent* pipeline:

tokenization → lemmatization → tagging → parsing

Challenges

- Errors propagate: mistakes at an early stage can affect later annotations.
- Reported accuracy may be optimistic: evaluation may assume that earlier processing steps are already correct.
- Overall accuracy can be misleading: high average accuracy may hide poor performance on specific tags or features.
- General NLP tools may not directly capture the features relevant to an SLA research question.

Kyle & Eguchi (2024) provides an example of how NLP accuracy can be evaluated for L2 English research.

Project: Narrowing your RQ

1. What do you want to understand?

- ☐ Learner language
- ☐ Language development
- ☐ Language teaching
- ☐ Language assessment
- ☐ Other: _____

2. What type of corpus/data?

- ☐ Informal writing
- ☐ Academic writing
- ☐ Spoken interviews
- ☐ Classroom interaction
- ☐ Assessment responses
- ☐ Discussion/forum data
- ☐ Social media
- ☐ Other: _____

3. What research design?

- ☐ Descriptive / single-corpus
- ☐ Cross-sectional
- ☐ Longitudinal
- ☐ Pseudo-longitudinal
- ☐ Comparative
- ☐ Other: _____

4. What metadata matter?

- ☐ L1
- ☐ Proficiency
- ☐ Age
- ☐ Genre
- ☐ Task conditions
- ☐ Other: _____

5. What linguistic feature/construct?

- ☐ Vocabulary
- ☐ Phraseology
- ☐ Grammar
- ☐ Accuracy
- ☐ Complexity
- ☐ Fluency
- ☐ Discourse organization
- ☐ Pragmatic / interactional features
- ☐ Other: _____

6. What NLP process/tool might you need?

- ☐ Tokenization
- ☐ Lemmatization
- ☐ POS tagging
- ☐ Dependency parsing
- ☐ Automatic classification
- ☐ Other: _____

7. How will you check NLP accuracy?

- ☐ Existing validation evidence
- ☐ Manual checking of a sample
- ☐ Comparison with human annotation
- ☐ Accuracy for the specific feature(s) I need

8. Where might human judgment be needed during/after NLP processing?

- ☐ Error identification
- ☐ Intended meaning / ambiguity
- ☐ Annotation decisions
- ☐ Interpretation of the feature/construct
- ☐ Other: _____